# Task 5

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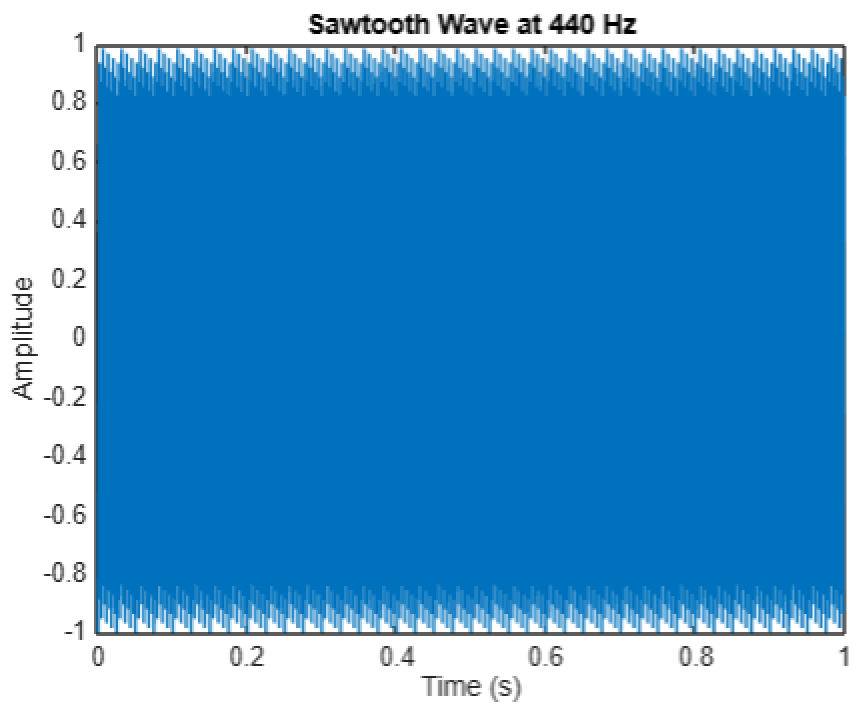
## A.

**Generate a 1-second sawtooth wave with a fundamental frequency of 440 Hz (the musical note 'A').**




The generated plot displays a 1-second sawtooth waveform with a fundamental frequency of 440 Hz, corresponding to the musical note 'A'. The waveform exhibits a linearly rising ramp followed by a sharp drop, repeating every 1/440 seconds. Over the duration of one second, a total of 440 such cycles are observed.

The amplitude oscillates between -1 and 1, illustrating the periodic and non-sinusoidal nature of the signal. This pattern lives up to its name — *sawtooth* — as it visually resembles the teeth of a saw.

The sharp discontinuities in the wave contribute to its rich harmonic structure. Due to the high frequency, the plot appears dense; zooming into a smaller interval would reveal the detailed shape of individual cycles.

## B.

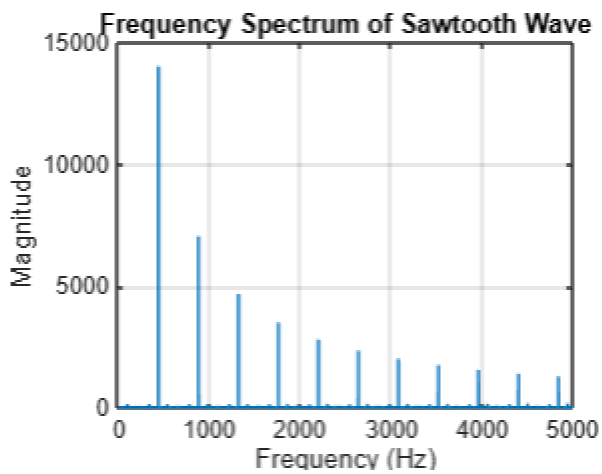
**Listen to the wave and compare its timbre (sound quality) to that of a pure sine wave.**

When listening to the sawtooth and sine waves, a clear difference in timbre is noticeable. The sawtooth wave sounds bright and harsh due to its rich harmonic content, while the sine wave sounds pure and soft with only a single frequency component. This comparison illustrates how different waveforms, despite having the same pitch (frequency), produce distinctly different sound qualities.

## C.

**Apply the Fourier Transform and plot the frequency spectrum. A sawtooth wave contains both even and odd harmonics. Identify the first few harmonics in your plot.**

This is the plot output:




The Fourier Transform decomposes the sawtooth wave into its frequency components. The sawtooth wave includes both **even and odd harmonics** of the fundamental frequency (440 Hz). In the frequency spectrum, we observe peaks at:

- 440 Hz (1st harmonic),
- 880 Hz (2nd harmonic),
- 1320 Hz (3rd harmonic),
- 1760 Hz (4th harmonic),
- and so on.

These harmonics gradually decrease in amplitude, creating the **rich and buzzy** timbre of the sawtooth wave.

## D.

**How does the presence of all integer harmonics in the sawtooth wave's spectrum explain its "buzzy" or "bright" sound compared to a sine wave?**

The sawtooth wave contains all the integer harmonics of its fundamental frequency — i.e., both odd and even harmonics:
f, 2f, 3f, 4f, …

Each one contributes more frequency content to the sound. The more harmonics in a waveform, the richer and brighter the sound.

A sine wave, on the other hand, contains just the fundamental frequency with no harmonics, producing a smooth and pure sound.

Since the sawtooth wave has many strong harmonics (especially at larger values of frequency), it's "buzzy" or "raspy," similar to a violin string or synthesizer. It has this rich frequency content and therefore has a bright timbre unlike the pure tone of a sine wave.